MATH MA'AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 2: COMPUTATION OPERATIONS

Learning objectives: Addition • Subtraction • Estimating Sum and Difference • Multiplication • Division • Estimating Products and Quotients • Factors and Multiples • H.C.F. and L.C.M.

2.1 ADDITION

Addition means combining two or more numbers. The numbers being added are called **addends**, and the answer is called the **sum**.

Example: 543,218 + 230,121 = 773,339

Properties of Addition

• **Commutative property:** Changing the order of addends does not change the sum. Example: 125 + 350 = 350 + 125.

• **Associative property:** Changing the grouping of addends does not change the sum. Example: (20 + 30) + 40 = 20 + (30 + 40).

• **Additive identity:** Adding zero to a number gives the same number. Example: 4,216 + 0 = 4,216.

• **Successor:** Adding 1 gives the next number. Example: 25,168 + 1 = 25,169.

• Adding 10, 100 or 1,000 increases the tens, hundreds or thousands place by one unit respectively, provided there is no carrying into another place.

Remember: Addition can be checked by reversing the order of the addends or by subtracting one addend from the sum.

2.2 SUBTRACTION

Subtraction means taking one number away from another. The number from which we subtract is the **minuend**; the number being subtracted is the **subtrahend**; the answer is the **difference**.

Example: 725,685 – 524,353 = 201,332

Properties and useful facts

• Subtracting zero leaves the number unchanged: 72,568 – 0 = 72,568.

• Subtracting a number from itself gives zero: 82,534 – 82,534 = 0.

• Subtracting 1 gives the predecessor: 549,876 – 1 = 549,875.

• Subtracting 10, 100 or 1,000 decreases the tens, hundreds or thousands place by one unit respectively, provided borrowing is not needed in other places.

• Subtraction is not commutative: 15 – 8 is not equal to 8 – 15.

Check: Difference + Subtrahend = Minuend. For example, 201,332 + 524,353 = 725,685.

2.3 ESTIMATING THE SUM AND DIFFERENCE

Estimation means finding an approximate answer. Round each number to the same suitable place value, then perform the operation.

Rounding reminder

• If the digit immediately to the right is 0–4, keep the rounding digit unchanged.

• If the digit immediately to the right is 5–9, increase the rounding digit by 1.

• Replace all digits to the right of the rounded place with zeros.

Calculation	Rounding	Estimated answer
75,396 + 45,256	75,000 + 45,000 (nearest 1,000)	1,20,000
75,396 – 45,256	75,000 – 45,000 (nearest 1,000)	30,000
48,765 + 32,148	49,000 + 32,000 (nearest 1,000)	81,000

Tip: Choose the rounding place according to the question. Estimation gives an approximate answer, not the exact answer.

2.4 MULTIPLICATION

Multiplication is repeated addition of equal groups. The numbers multiplied are called **factors**, and the answer is called the **product**.

Example: 125 × 24 = 125 × (20 + 4) = 2,500 + 500 = 3,000.

Properties of Multiplication

• **Multiplicative identity:** Any number multiplied by 1 remains unchanged. Example: 4,235 × 1 = 4,235.

• **Zero property:** Any number multiplied by 0 gives 0.

• **Commutative property:** Changing the order of factors does not change the product. Example: 12 × 8 = 8 × 12.

• **Associative property:** Changing the grouping of factors does not change the product. Example: (2 × 3) × 4 = 2 × (3 × 4).

• **Multiplication by 10, 100 and 1,000:** The product is 10, 100 and 1,000 times the original number respectively.

Examples: 94,567 × 10 = 945,670
94,567 × 100 = 9,456,700
94,567 × 1,000 = 94,567,000

Useful method: Break a number into tens, hundreds or thousands to make multiplication easier. Example: 36 × 45 = 36 × (40 + 5) = 1,440 + 180 = 1,620.

2.5 DIVISION

Division means sharing a quantity equally or making equal groups. The number being divided is the **dividend**, the number dividing it is the **divisor**, and the answer is the **quotient**.

Example: 84,672 ÷ 24 = 3,528

Properties and important rules

• Dividing a number by 1 gives the number itself: 65,241 ÷ 1 = 65,241.

• Zero divided by a non-zero number is zero: 0 ÷ 25 = 0.

• A non-zero number divided by itself is 1: 75,042 ÷ 75,042 = 1.

• **Division by zero is not defined.** A number cannot be divided by zero.

• Division is not commutative: 20 ÷ 5 is not equal to 5 ÷ 20.

Division with remainder

When a number cannot be divided exactly, there may be a remainder. The remainder is always smaller than the divisor.

Important formula:
Dividend = Divisor × Quotient + Remainder
Example: 35 × 125 + 18 = 4,393. Therefore, 4,393 ÷ 35 gives quotient 125 and remainder 18.

2.6 ESTIMATING MULTIPLICATION AND DIVISION

To estimate a product or quotient, round the numbers to convenient values before calculating. The aim is a quick, sensible approximation.

Estimating multiplication

Example: 503 × 4,345
Round 503 to 500 and 4,345 to 4,000.
Estimated product = 500 × 4,000 = 20,00,000.

Estimating division

Example: 6,120 ÷ 29
Use nearby convenient numbers: 6,000 ÷ 30 = 200.
Estimated quotient ≈ 200.

Tip: Round to numbers that are easy to multiply or divide. Different reasonable rounding choices may give slightly different estimates.

2.7 FACTORS AND MULTIPLES

Factors

A **factor** of a number divides it exactly, leaving no remainder. Factors are finite in number.

Example: Factors of 12 are 1, 2, 3, 4, 6 and 12.

Properties of Factors

• 1 is a factor of every whole number.

• Every number is a factor of itself.

• The smallest factor of a positive whole number is 1.

• The greatest factor of a positive whole number is the number itself.

• Every positive whole number has at least two factors, except 1, which has only one factor.

Prime and Composite Numbers

Type	Meaning	Examples
Prime	A whole number greater than 1 with exactly two factors: 1 and itself.	2, 3, 5, 7, 11
Composite	A whole number greater than 1 with more than two factors.	4, 6, 8, 9, 10
Special case	1 is neither prime nor composite.	1

Olympiad facts: 2 is the only even prime number. Every prime number greater than 2 is odd. The number 1 is neither prime nor composite.

Multiples

A **multiple** is obtained by multiplying a number by a whole number. Multiples continue without end.

Example: Multiples of 5 are 5, 10, 15, 20, 25, ...

• Every number is a multiple of itself.

• Every number is a multiple of 1.

• A number has infinitely many positive multiples.

• The smallest positive multiple of a number is the number itself.

Common factors: Factors shared by two or more numbers.
Common multiples: Multiples shared by two or more numbers.

H.C.F. – HIGHEST COMMON FACTOR

The **H.C.F.** of two or more numbers is the greatest number that divides each of them exactly. It is also called the greatest common divisor (G.C.D.).

Example: Factors of 12 = 1, 2, 3, 4, 6, 12
Factors of 18 = 1, 2, 3, 6, 9, 18
Common factors = 1, 2, 3, 6
Therefore, H.C.F. (12, 18) = 6.

Prime factorisation method

Write each number as a product of prime factors. Select only the prime factors common to all numbers, using the smallest power of each. Multiply them to get the H.C.F.

Example: 12 = 2 × 2 × 3; 18 = 2 × 3 × 3. Common prime factors are 2 and 3. H.C.F. = 2 × 3 = 6.

L.C.M. – LEAST COMMON MULTIPLE

The **L.C.M.** of two or more numbers is the smallest positive number that is exactly divisible by all the given numbers.

Example: Multiples of 5 = 5, 10, 15, 20, 25, 30, ...
Multiples of 6 = 6, 12, 18, 24, 30, ...
The smallest common positive multiple is 30. Therefore, L.C.M. (5, 6) = 30.

Prime factorisation method

Write each number as a product of prime factors. Select every prime factor with its greatest power appearing in any factorisation, then multiply.

Example: 12 = 2^2 × 3; 14 = 2 × 7. L.C.M. = 2^2 × 3 × 7 = 84.

For two positive whole numbers:
H.C.F. × L.C.M. = Product of the two numbers.
Example: For 12 and 18: 6 × 36 = 12 × 18 = 216.

Remember: H.C.F. cannot be greater than the smallest given number. L.C.M. is always a multiple of each given number.

QUICK REVISION – IMPORTANT FORMULAE

Concept	Formula / Rule
Addition	Addends + Addends = Sum
Subtraction	Minuend – Subtrahend = Difference
Multiplication	Factor × Factor = Product
Division	Dividend ÷ Divisor = Quotient (if exact)
Division with remainder	Dividend = Divisor × Quotient + Remainder
H.C.F.	Greatest common factor
L.C.M.	Smallest positive common multiple
Two-number relation	H.C.F. × L.C.M. = Product of the numbers

IMO preparation checklist: Practise carrying and borrowing in large-number operations, properties of operations, estimation, division with remainders, prime/composite classification, factor lists, prime factorisation, H.C.F. and L.C.M. word problems.

ALL THE BEST